FURTHER MECHANICS / CIRCULAR MOTION & OSCILLATIONS

# Motion in a Horizontal Circle

Questions by topic 15 questions · 146 marks

EXAM BOARDS Edexcel FM2 9FM0/4C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Major Y421 CIE 9231 Paper 3

**Instructions**

Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question gives a value. Where a question leaves $g$ unspecified and a final answer depends on it, the solutions also give that answer for $g = 10\,\mathrm{m\,s^{-2}}$.

CIE (9231 Paper 3): papers take $g = 10\,\mathrm{m\,s^{-2}}$; use the $g = 10$ final answers where the solutions give them.

## Questions


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|---|---|---|
| Question 1 | *10 marks* | 2 |
| Question 2 | *11 marks* | 3 |
| Question 3 | *9 marks* | 4 |
| Question 4 | *11 marks* | 5 |
| Question 5 | *12 marks* | 6 |
| Question 6 | *8 marks* | 7 |
| Question 7 | *7 marks* | 8 |
| Question 8 | *7 marks* | 9 |
| Question 9 | *7 marks* | 10 |
| Question 10 | *9 marks* | 11 |
| Question 11 | *11 marks* | 12 |
| Question 12 | *12 marks* | 13 |
| Question 13 | *10 marks* | 14 |
| Question 14 | *11 marks* | 15 |
| Question 15 | *11 marks* | 16 |

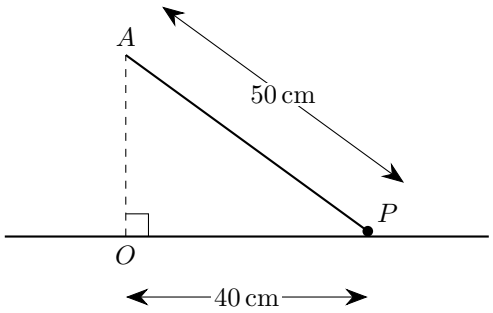
**1.** A cyclist rides round a circular track which is banked at a constant angle to the horizontal.
The cyclist and bicycle are modelled as a particle moving at constant speed in a horizontal circle, taking 22 s to complete each lap.
At this speed there is no sideways frictional force on the bicycle, and the magnitude of the normal reaction of the track on the particle is $\frac{25}{24}$ times the weight of the particle.

**(a)** Find the radius of the circle. **[7]**

**(b)** Identify one limitation of this model. **[1]**

The cyclist now slows down, so that each lap of the same horizontal circle takes 30 s, again at constant speed. The cyclist and bicycle are still modelled as a particle.

**(c)** Describe the extra force that will now be acting on the particle, stating its direction and giving a reason for this direction. **[2]**



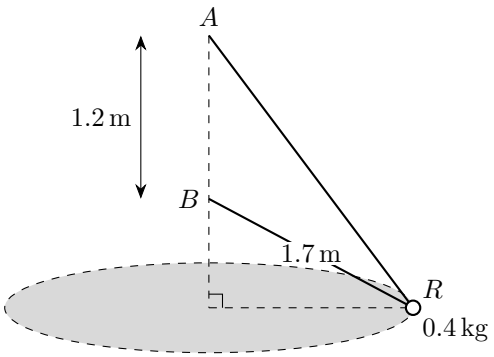

**2.** A particle $P$ of mass 0.8 kg is attached to one end of a light inextensible string of length 50 cm. The other end of the string is attached to a fixed point $A$ that is vertically above the point $O$ on a smooth horizontal table.
The particle remains in contact with the table, with the string taut, and moves with constant speed in a horizontal circle with centre $O$ and radius 40 cm, as shown in the diagram above.
The tension in the string and the normal reaction between $P$ and the table are equal in magnitude.

**(a)** (i) Find the tension in the string. **[3]**

(ii) Find the speed of $P$. **[3]**

The string will break if the tension in it exceeds 15 N. The speed of $P$ is now gradually increased.

**(b)** Show that $P$ loses contact with the table before the string breaks, and find the speed of $P$ at the instant it loses contact. **[5]**




**3.** The diagram above shows a light inextensible string of length 4.2 m passing through a small smooth ring $R$ of mass 0.4 kg. The ends of the string are attached to fixed points $A$ and $B$, where $A$ is 1.2 m vertically above $B$.
The ring moves with constant speed in a horizontal circle whose centre lies on the line through $A$ and $B$, vertically below $B$. The distance $BR$ is 1.7 m and both sections of the string are taut.

**(a)** Show that the radius of the circle is 1.5 m. **[2]**

**(b)** Find the tension in the string. **[3]**

**(c)** Find the time taken for $R$ to complete one revolution. **[4]**

**4.** A particle moves with constant speed on a circular path of radius $2\sqrt{2}\,\text{m}$ in a horizontal plane.
The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in this plane.
The centre of the circle has position vector $(3\mathbf{i} + 3\mathbf{j})\,\text{m}$ relative to the origin.
At time $t = 0$, the particle is at the point with position vector $(5\mathbf{i} + \mathbf{j})\,\text{m}$ and its velocity is $-\frac{\pi}{2}(\mathbf{i} + \mathbf{j})\,\text{m s}^{-1}$.

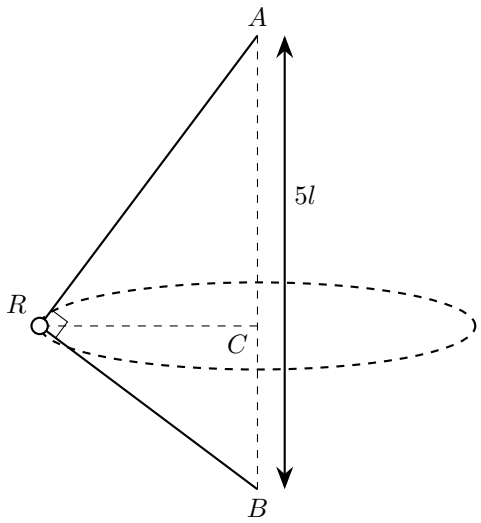
**(a)** Calculate the angular speed of the particle. **[2]**

**(b)** Find an expression for the position vector of the particle at time $t$ seconds. **[3]**

**(c)** Find an expression for the acceleration of the particle at time $t$ seconds. **[3]**

**(d)** State the magnitude of the acceleration of the particle. **[1]**

**(e)** Find the time when the acceleration of the particle is first directed towards the origin. **[2]**

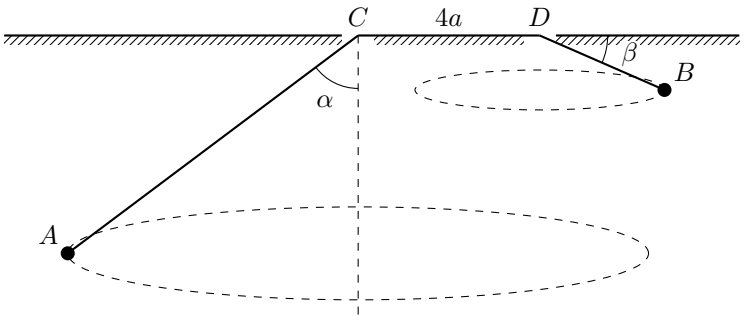



**5.** One end of a light inextensible string of length $7l$ is attached to a fixed point $A$. A small smooth ring $R$ of mass $m$ is threaded on the string, and the other end of the string is attached to a fixed point $B$, vertically below $A$, with $AB = 5l$.
The ring moves with constant speed in a horizontal circle whose centre $C$ lies on $AB$. Both sections of the string are taut and angle $ARB = 90°$, as shown in the diagram above.

**(a)** Show that $AR$ is either $3l$ or $4l$. **[2]**

**(b)** By resolving vertically for the ring, show that $AR = 4l$ and find, in terms of $m$ and $g$, the tension in the string. **[5]**

**(c)** Find, in terms of $g$ and $l$, the time taken for the ring to complete one revolution. **[5]**

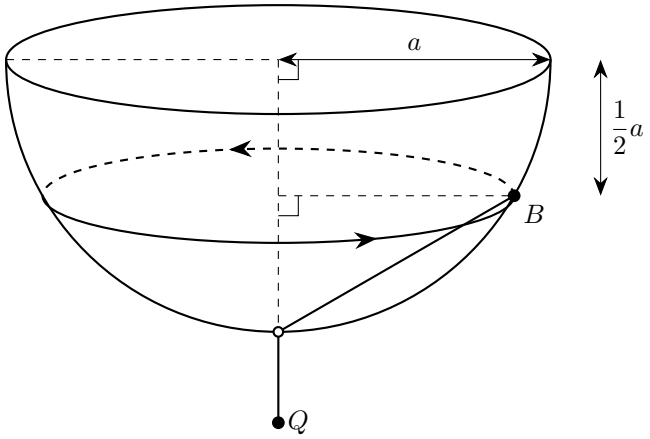



**6.** A light inextensible string $AB$ of length $15a$ passes through two small holes $C$ and $D$ in a smooth horizontal table, where $CD = 4a$.
A particle of mass $3m$ is attached at the end $A$ and moves in a horizontal circle, centred vertically below $C$, with constant angular speed $\omega$.
A particle of mass $2m$ is attached at the end $B$ and moves in a horizontal circle, centred vertically below $D$, with constant angular speed $2\omega$.
$AC$ makes an angle $\alpha$ with the downward vertical, where $\cos\alpha = \frac{3}{5}$, and $DB$ makes an angle $\beta$ with the horizontal, as shown in the diagram above.

**(a)** Find the lengths $AC$ and $DB$ in terms of $a$. **[4]**

**(b)** Find $\omega$ in terms of $a$ and $g$, and find the vertical distance of $B$ below the table in terms of $a$. **[4]**

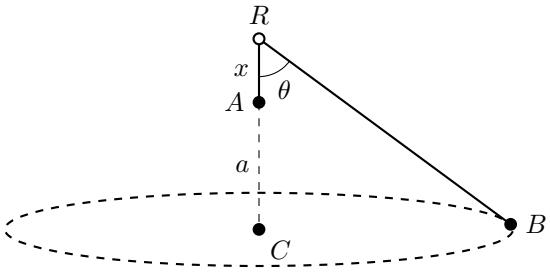
**7.** A hollow hemispherical bowl of radius $a$ has a rough inner surface and is fixed with its axis vertical.
A particle $P$ of mass $m$ moves with constant speed $\sqrt{\frac{6}{5}ga}$ in a horizontal circle on the inner surface of the bowl, at a height $\frac{1}{5}a$ above the lowest point of the bowl.
The coefficient of friction between $P$ and the bowl is $\mu$, and $P$ is on the point of slipping up the surface of the bowl.
Find the value of $\mu$. **[7]**




**8.** A hemispherical shell of radius $a$ is fixed with its rim uppermost and horizontal. There is a small smooth hole at the lowest point of the shell.
A small bead, $B$, of mass $m$, is moving with constant angular speed, $\omega$, in a horizontal circle on the smooth inner surface of the shell. The centre of the path of $B$ is at a distance $\frac{1}{2}a$ vertically below the level of the rim of the hemisphere, as shown in the diagram above.
A light inextensible string is attached to $B$ and passes through the hole. A particle, $Q$, of mass $2m$ is attached to the other end of the string and hangs at rest vertically below the hole.

**(a)** Find, in terms of $m$ and $g$, the magnitude of the normal reaction of the shell on $B$. **[4]**

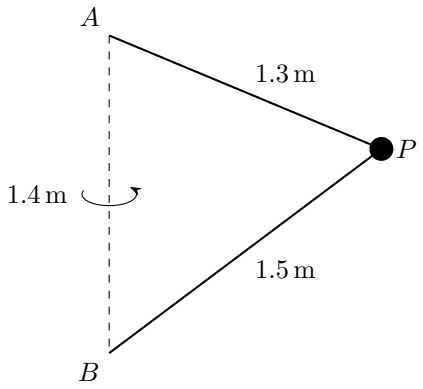
**(b)** Find the magnitude of $\omega$, giving your answer in terms of $a$ and $g$. **[3]**




**9.** A light inextensible string of length $3a$ is threaded through a fixed smooth ring $R$.
One end of the string is attached to a particle $A$ of mass $5m$, and the other end is attached to a particle $B$ of mass $3m$.
The particle $A$ hangs in equilibrium at a distance $x$ vertically below the ring, and the angle between $AR$ and $BR$ is $\theta$, as shown in the diagram above.
The particle $B$ moves with constant speed in a horizontal circle whose centre $C$ is a distance $a$ vertically below $A$.

**(a)** Show that $\cos\theta = \frac{3}{5}$ and find $x$ in terms of $a$. **[4]**

**(b)** Find the speed of $B$ in terms of $a$ and $g$. **[3]**




**10.** A particle $P$ of mass 0.7 kg is attached to one end of a light inextensible string of length 1.3 m. The other end of this string is attached to the fixed point $A$.
A second light inextensible string, of length 1.5 m, has one end attached to $P$ and the other end attached to the fixed point $B$. The point $B$ is vertically below $A$ with $AB = 1.4$ m, as shown in the diagram above. The particle moves in a horizontal circle with constant angular speed $\omega$ rad s$^{-1}$, with both strings taut.

**(a)** Show that the tension in the string $BP$ is $\frac{3}{8}\left(\omega^2 - 2g\right)$ N, where $g$ m s$^{-2}$ is the acceleration due to gravity. **[7]**

**(b)** The string $BP$ will break if the tension in it exceeds 15 N.
Hence find the range of values of $\omega$ for which the tension in $BP$ is positive and $BP$ does not break. **[2]**

**11.** A light inextensible string has length $35a$. One end of the string is attached to a fixed point $A$ and the other end is attached to a fixed point $B$, where $A$ is vertically above $B$ and $AB = 7a$.
A small ball of mass $m$ is attached to a point $P$ on the string, where $AP = 20a$.
The ball moves in a horizontal circle with constant speed $v$, with both $AP$ and $BP$ taut.
The string will break if the tension in either part of it exceeds $\frac{10mg}{7}$.
Model the ball as a particle and assume that the string does not break.

**(a)** Show that

$$9ag < v^2 \leq 15ag$$ **[8]**

**(b)** Find, in terms of $a$ and $g$, the time taken for the ball to make one complete revolution when the tensions in $AP$ and $BP$ are equal. **[3]**

**12.** In this question use $g = 9.81\,\mathrm{m\,s^{-2}}$.
A conical pendulum is made from an elastic string and a sphere of mass $0.5\,\mathrm{kg}$.
The string has natural length $2\,\mathrm{m}$ and modulus of elasticity $100\,\mathrm{N}$.
The sphere describes a horizontal circle at a constant speed of $5\,\mathrm{m\,s^{-1}}$.
The angle between the elastic string and the vertical is $\alpha$.

**(a)** Show that

$$25\cos^2\alpha = 9.81\sin^2\alpha\,(2\cos\alpha + 0.0981)$$ **[7]**

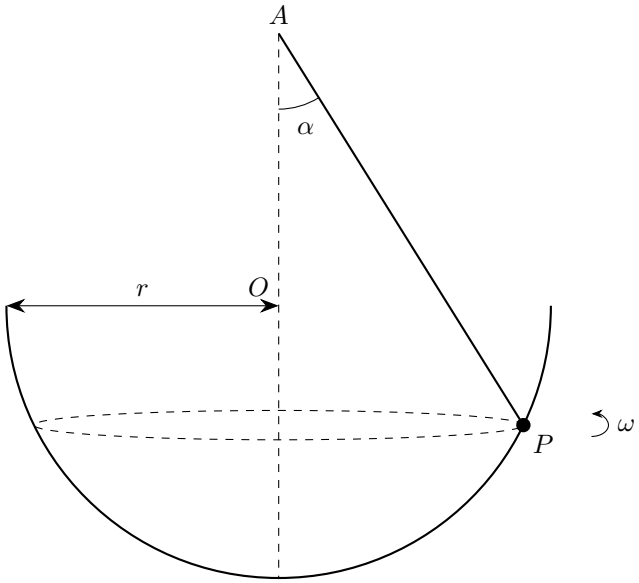
**(b)** Use your calculator to find $\alpha$, giving your answer in degrees. **[1]**

**(c)** Find the elastic potential energy stored in the string. **[4]**

**13.** A hollow right circular cone $C$ of height 3 m and base radius 4 m is fixed with its axis vertical and its vertex downwards.
One end of a light elastic string of natural length 3 m and modulus of elasticity 16 N is fixed to the vertex of $C$. The other end of the string is attached to a particle $P$ of mass 5 kg.
$P$ moves in a horizontal circle with constant speed and in contact with the smooth inner surface of $C$.

**(a)** Find the greatest speed of $P$ for which the tension in the string is zero. **[3]**

**(b)** The speed of $P$ is now $6\,\mathrm{m\,s^{-1}}$.
Find the extension of the string and the magnitude of the normal reaction between $P$ and $C$. **[7]**




**14.** The diagram above shows a hemispherical bowl. The rim of this bowl is a circle with centre $O$ and radius $r$. The bowl is fixed with its rim horizontal and uppermost.
The point $A$ is fixed vertically above $O$, with $OA = r$. A particle $P$, of mass $m$, is connected to $A$ by a light inextensible string and describes a horizontal circle with constant angular speed $\omega$ on the smooth inner surface of the bowl.
The string is taut, and $AP$ makes an angle $\alpha$ with the vertical, where $0 < \alpha < \frac{\pi}{4}$.

**(a)** Show that the tension in the string is $2m\cos\alpha\left(g - r\omega^2\cos 2\alpha\right)$. **[7]**

**(b)** Find, in terms of $g$, $r$ and $\alpha$, the range of values of $\omega^2$ for which this motion is possible, with the string taut and $P$ in contact with the bowl. **[4]**

**15.** A horizontal turntable rotates with constant angular speed $\omega\,\text{rad}\,\text{s}^{-1}$ about a fixed vertical axis through its centre $O$.
One end of a light elastic string of natural length $l\,\text{m}$ and modulus of elasticity $\lambda\,\text{N}$ is attached to $O$. The other end of the string is attached to a particle $A$ of mass $m\,\text{kg}$, which rests on the turntable.
The coefficient of friction between $A$ and the turntable is $\mu$.
The particle $A$ moves with the turntable, with the string horizontal and taut, and $A$ is on the point of slipping away from $O$. The extension of the string is denoted by $x\,\text{m}$.

**(a)** Show that

$$x = \frac{ml\left(l\omega^2 - \mu g\right)}{\lambda - ml\omega^2} \qquad \textbf{[3]}$$

**(b)** Using a binomial series, show that if $\lambda$ is very large then

$$\lambda x \approx ml\left(l\omega^2 - \mu g\right)\left(1 + \frac{ml\omega^2}{\lambda}\right) \approx ml\left(l\omega^2 - \mu g\right) \qquad \textbf{[3]}$$

**(c)** By finding an approximation for the tension in the string, explain how the result obtained when $\lambda$ is very large relates to the situation when the string is inextensible and of length $l\,\text{m}$. **[1]**

The nature of the surface of the turntable is such that the modelling assumption that $\mu$ is constant is justifiable provided that the angular speed does not exceed $12\,\text{rad}\,\text{s}^{-1}$.
In the case where $m = 0.25$, $l = 0.5$, $\lambda = 400$ and $\mu = 0.4$, the extension of the string is measured as being $2.0\,\text{cm}$.

**(d)** Use the approximation $\lambda x \approx ml\left(l\omega^2 - \mu g\right)$ to estimate the value of $\omega$. **[2]**

**(e)** By comparing the radius used in part **(d)** with the true radius of the circle, explain whether your estimate is greater or less than the true value of $\omega$, and hence whether the modelling assumption is necessarily justifiable in this situation. **[2]**